TECHNICAL ANALYSIS OF THE BATTLEFIELD SURVEILLANCE SYSTEM (PROJECT SANJAY) AND NETWORK-CENTRIC WARFARE

Bharat Electronics Limited

Domain - Defence

B.Tech CSE (Specialization in Cyber Security)
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01 February, 2026

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2. Executive Summary

- **Bharat Electronics Limited** is an Indian public sector, primarily an aerospace and defence electronics company, headquartered in Bangalore. It primarily manufactures advanced electronic products for ground and aerospace applications. BEL is one of sixteen PSUs under the administration of the Ministry of Defence. It has been granted Navratna status by the Government of India.
- BEL's primary focus is on producing futuristic defence electronics, with major of its revenue derived from the defence sector. Its core areas include:

I. **Missile Systems:** Design and integration of air defence systems, including the Akash missile system.

II. **Communication & Network Systems:** Software Defined Radios, secure communications, and C4I systems.

- **Key Technical Innovations**: **Akash Air Defence System -** BEL helped in the integration of this indigenous system, over 95% material sourced domestically.
- **Major Findings & Strategic Developments: "Atmanirbhar Bharat" Focus -** 77-78% of turnover in recent years came from indigenous technology.
- **Strategic Partnerships:** Formed joint ventures like BEL IAI AeroSystems (for MRSAM) and BEL-Thales Systems, and partnerships with startups for UAV technologies.

3. **Company Overview**

- **Organization name -** Bharat Electronics Limited & Indian Army
- **Year of establishment -** 1954 (BEL) / BSS launched Jan 2025

  (Development began early 2000s)
- **Founders -** Defence Research and Development Organisation (DRDO),

  Centre for Artificial Intelligence and Robotics Lab & BEL
- **Headquarters -** Bengaluru, Karnataka (BEL & CAIR); New Delhi (Army HQ)
- **Global presence –** Primarily India
- **Company Size:**

Revenue: ₹23,000 Crore

Employees: 8,937+ permanent employees

User Base: 12 Lakh+ active personnel (Indian Army)

4. Industry & Domain Analysis

- **Industry Segment:** Aerospace and defence electronics.
- **Specific Focus:** Battlefield Management Systems and Data Fusion for Decision making.
- **Market Position: BEL** has a near-monopoly in strategic defence electronics in India, accounting for approximately 40% of the industry's total sales and 45% of its market capitalization.

Key Competitors:

- **Domestic:** Tata Advanced Systems Ltd (TASL) and Larsen & Toubro (L&T) Defence, competing in radar, missile systems, and overall C4I integration.
- **Global (C4I Systems):**
  - **Raytheon & Lockheed Martin (USA):** Global leaders in the missile defence and C2 (Command & Control) systems.
  - **Thales Group (France) & Elbit Systems (Israel):** Leading providers of the tactical communication and surveillance solution to India.
  - **BAE Systems (UK):** Major competitor in encrypted military communications and hardware.

Industry Challenges:

- **Integration Complexity:** The biggest barrier is the "System of Systems" challenge—integrating modern IP-based network with old radios/tanks effectively.
- **Cybersecurity Threats:** Protecting from Advanced Persistent Threats (APTs) and Electronic Warfare i.e., jamming is increasingly difficult as systems become more digitized.

5. Problem Statement / Business Challenge

- **The Challenge:** The Indian Army faced "Information Isolation" where commanders could not see a unified picture of the battlefield in real-time.
- **Software Goal:** To create a centralized software platform that fuses data from sensors, satellites, and ground units into "Command Operation" to speed up decision-making cycles.

6. Technical Architecture

  - Mobile Automated Surveillance System
  - The system consists of a monitoring centre, a communication control unit, a generator and various monitoring communication terminals.
  - It collects data from various types of sensors located in the field
  - A map displays the strategic position on the background
  - Provides decision support to commanders using multi-sensor data fusion techniques
  - It uses artificial intelligence and information-based techniques to identify and assess the position of the target
  - Voice, video, content and images are sent over highly secure high speed digital communication links
  - Fibre optics and VHF, HF and UHF radio, hybrid communication networks have multi-layered security.

- **Technology Stack:**
  - **Operating System:** Maya OS (a hardened operating system, Ubuntu distribution of Linux).
  - **Database:** Relational Database Management Systems.
  - **Framework:** Gati Shakti GIS (Geographic Information System) based user interface.
- **APIs & Integrations:**
  - **Integration:** The system functions as a central hub connecting multiple subsystems like **ACCCS** (Artillery Combat Command Control System) and

CIDSS (Combined Information Decision Support System) to create a unified view.

- **Sensor Integration:** Direct hardware-level integration with **Swathi Weapon Locating Radars** and **LORROS** (Long Range Reconnaissance Observation Systems) via serial and IP-based tactical interfaces.

- **Data Collection:**
- Sources: Ingests large volumes of ISR data from Heron UAVs, SATCOM (Satellite Communication), and Thermal Imagers.
- Input Method: A combination of automated sensor inputs (IP-based) and manual entry from soldier-portable "Hand Held Computers" at the battalion level.

7. Innovation & Competitive Advantage

- **Multi-Sensor Data Fusion:** Unlike traditional systems that needed manual correlation of reports, this system automatically combines data from UAVs, Satellites, and Ground Radars into a single "Common Operational Picture" (COP). It employs algorithms to check authenticity and eliminate duplicate targets in real-time.
- **Sensor-to-Shooter Integration:** The system establishes a digital closed loop where a target identified by a sensor is immediately available to a shooter through the **ACCCS** integration, significantly reducing the "Kill Chain" from minutes to seconds.
- **Indigenous Architecture:** The system employs an indigenous "Chakravyuh", an endpoint detection and protection system and Maya-OS to safeguard against malware and foreign cyber-espionage, a key innovation over imported defence software.
- **Gati Shakti-Inspired GIS**: It employs a "System of Systems" approach inspired by the PM Gati Shakti platform, overlaying logistics, terrain, and weather information on a single GIS screen for multi-domain situational awareness.
- **Information Dominance**: With its integrated battlefield view, the system removes delay for Indian commanders, enabling them to act quicker than the enemy (OODA Loop advantage).

8. Use Cases & Applications

- **Real-world Use Cases:**
    - **Border Surveillance & Intrusion Detection:** The system is actively deployed along the Line of Control (LoC). It aggregates feeds from thermal imagers and LORROS sensors to alert commanders of suspicious movements in "dead ground" (areas not visible to the naked eye) during night patrols.
- **Industries Served:**
- **Primary:** Defence & National Security.
- **Secondary:** Aerospace (UAV integration) and Telecommunications (Secure tactical networks).

9. Performance & Impact

- **Performance Metrics:**
- **Deployment Scale:** The project has been cleared for induction across all operational Corps in three phases (March – October 2025) with a budget of roughly ₹2,402 Crore.
- **Efficiency Gains:** Trial & testing in the Thar desert showed a significant reduction in the OODA (Observe-Orient-Decide-Act) loop, allowing Indian forces to react to enemy movement nearly 40% faster than with voice-radio methods.
- **Market Impact:**
- **Indigenization:** Shift from imported Israeli command systems to the indigenous BEL-developed platform, the Army has retained critical data sovereignty and reduced long-term maintenance costs by relying on domestic vendors.

10. Limitations & Risks

- **Integration with aged Hardware**: One of the biggest challenges has been the backward compatibility problem. Integrating current IP-based software with 1980s analog radio communication systems in older tanks is very challenging and has often led to data latency.
- **Bandwidth Limitations:** In mountainous terrain such as Siachen, satellite bandwidth is limited. The large amount of data involved in sending high-resolution video from drones sometimes overload the network, causing delays in the tactical display.
- **Extra cost:** It has taken two decades to develop, and the cost has raised because of inflation and changes in the scope of work.
- **Supply Chain Risk:** Although the software is developed domestically, there is a risk of hardware backdoors because some hardware components (chips/processors) are imported.

11. Future Scope & Roadmap

- **Project Anumaan:** The future plan is to integrate with Project Anumaan, which will fetch micro-weather information from the National Centre for Medium-Range Weather Forecasting (NCMRWF) and display it on the commander's dashboard to improve artillery fire accuracy.
- **Project Avgat:** The plan is to integrate this system with Project Avgat, a Gati-Shakti-based initiative, to display logistics and supply chain information on the same map, ensuring that the commander is aware of the exact amount of ammunition available in the vicinity.
- **AI/ML Predictive Analysis: DRDO** & **CAIR** is developing an AI-based layer that not only shows the location of the enemy but also predicts what their next course of action would be based on terrain analysis and past doctrine.

12. Ethical, Legal & Social Implications

- **Officer Decision:** Contrary to the global trend of developing "killer robots," the Indian defence policy requires that while the software may detect the target, only a human officer can give the final call on a lethal strike.
- **Data Sovereignty:** As the system deals with geo-spatial data of the Indian territory, it comes under the purview of the Official Secrets Act (OSA). Legal provisions have been made to ensure that no cloud data is stored outside Indian servers, thus eliminating the possibility of foreign eavesdropping.
- **Soldier Safety:** As the system enhances situational awareness, it directly minimizes the chances of friendly fire and ambushes, which could save lives in conflict zones.

13. Conclusion

- The Command Information Decision Support System (CIDSS), developed into Project Sanjay, is a major breakthrough for the Indian Army in transitioning from Manual Warfare to Network-Centric Warfare. It addresses the age-old issue of "information isolation" by integrating a common digital battlefield.

- **Assessment:** When this project is assessed using the Standish Group Chaos Report criteria, it falls into the category of a Resolution Type 2 (Challenged) project. The project is currently operational and successful, but it was delivered later than scheduled and with substantial changes in budget.

- **Key Takeaway:** The success of this project, although delayed, proves that the "Growing Software" approach—beginning with a core system and then adding modules (as done recently in GIS improvements)—is better than trying to develop a perfect system in one attempt.

14. Appendix

Appendix A: System Deployment Photographs

Photo-1: Flag-off ceremony of Project SANJAY (Battlefield Surveillance System), indicating official induction into Indian Army operational units.

(Source: PIB, January 2025)

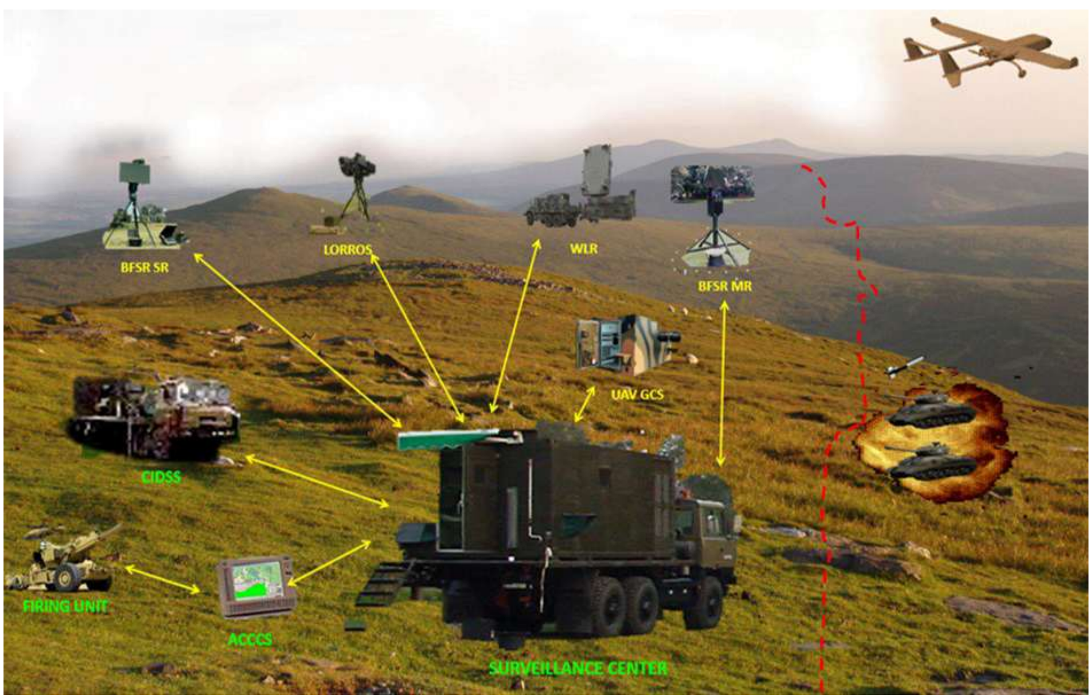

Photo-2: BSS collects data from various sensors to carry out situation assessment using MSDF and displays the tactical picture on a map background using a customized GIS.

(Source: BEL-India, 25 January 2025)

Appendix B: Battlefield Surveillance System – Data Flow Diagram

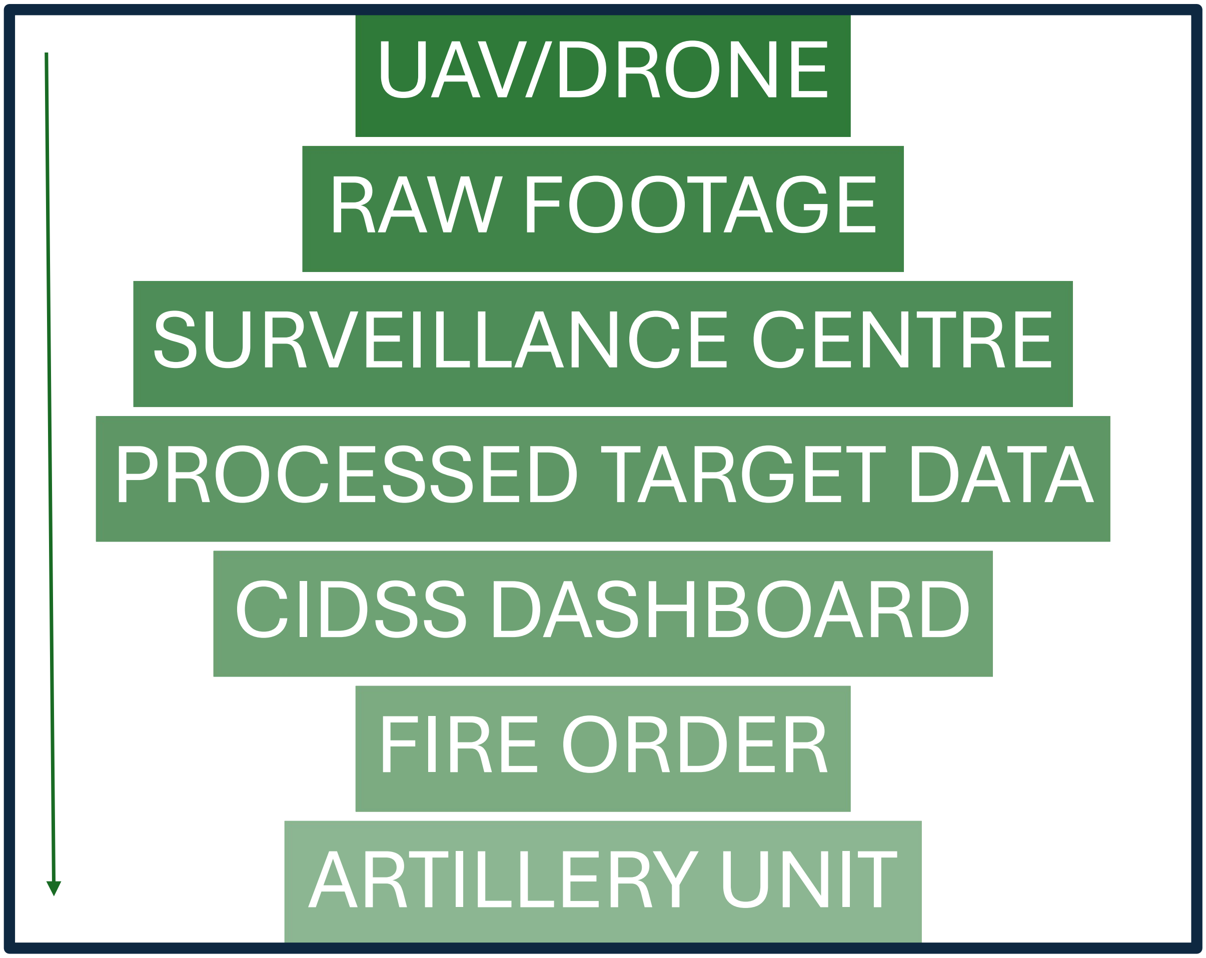

Fig-1: The overall data flow architecture of Project SANJAY

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